

Sahana Rayan

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ML researcher developing methods to evaluate and improve the reliability of AI systems, with research spanning model evaluation, uncertainty quantification, learning to defer, and foundation models. Experienced in developing and evaluating ML systems across language modeling, autonomous driving, and scientific ML.

Education

University of Michigan, Ann Arbor, MI

Doctor of Philosophy (Ph.D.) – Statistics

Master of Arts (M.A.) – Statistics

GPA: 3.97/4.00

Expected May 2027

2022 – 2024

- Awards: Ph.D. Student Service Award (2024), Honorable Mention - Outstanding GSI Team (2023)

Purdue University, West Lafayette, IN

Bachelor of Science (B.S.) – Computer Science

Bachelor of Science (B.S.) – Applied Statistics

GPA: 3.96/4.00

Aug 2018 – May 2022

- Awards: Graduated with Honors and Distinction (2022), Dean's List (2018 – 2022), College of Science Scholarship (2019), David S. Moore Scholarship (2019), Trustee's Scholarship (2018)

Industry Experience

Waymo

Jun 2026 – Sept 2026

Data Science Research Intern

- Developed a framework to evaluate Gemini's ability to triage autonomous-driving perception false negatives, estimating true false-negative likelihood from model predictions and structured reasoning signals.
- Designed and implemented an optimized sampling policy that translates Gemini's estimated false-negative likelihoods into sampling probabilities for downstream triage.
- Built an inverse-probability-weighted counterfactual evaluation framework to benchmark the proposed sampling policy against the existing sampling procedure.
- Increased estimated true false-negative discovery by ~20% over the existing sampling procedure.

Selected Research Experience

University of Michigan, Ann Arbor, MI

Aug 2022 – Present

Doctoral Researcher (Advised by Dr. Ambuj Tewari)

- *Learning to Partially Defer for Sequences (First Author)*
 - o Designed a learning-to-defer framework that identifies uncertain parts of long sequence outputs and selectively delegates them to stronger models or human experts, reducing costly full-output deferrals.
 - o Improved cost-accuracy trade-offs by up to 20% on news summarization LLMs, Traveling Salesman solvers, and autoregressive weather models, using PyTorch and Hugging Face Transformers.
 - o Awarded Best Oral Presentation - Michigan Student Symposium for Interdisciplinary Statistical Sciences (2025); submitted to TMLR.
- *Reasoning Models for Protein Disorder Prediction (First Author)*
 - o Trained Qwen3-4B to reason over heterogeneous biological evidence and predict intrinsically disordered protein regions, using GPT-5.4-generated reasoning traces for supervised fine-tuning and ESM-2 protein representations as soft tokens.

- o Post-trained the model with Group Relative Policy Optimization, designing rewards for prediction accuracy, structured output, and reasoning consistency; improved macro residue-level F1 from 0.65 after SFT to 0.85 after RL.
 - o Developed automated and LLM-as-judge evaluation pipelines for model-generated reasoning traces, using ROSCOE to assess reasoning coherence and an LLM judge to evaluate evidence faithfulness and hallucinations; found 78% of substantive biological claims grounded in the provided evidence; submitted to ICLR 2027.
- *Greedy Hybrid Partial Differential Equation (PDE) Solver (First Author)*
 - o Developed an approximate greedy router for hybrid iterative PDE solvers, dynamically selecting from an ensemble of classical numerical solvers and neural operators to accelerate PDE solving.
 - o Reduced time-to-solution by up to 75% over existing hybrid methods (e.g., HINTS) on Poisson, convection-diffusion, and anisotropic-diffusion equations; submitted to ICLR 2027.
- *Conformal Prediction for Robust Decision-Making (Second Author)*
 - o Engineered a robust optimization framework using conformal prediction to enhance decision-making under uncertainty in contextual optimization and linear control tasks.
 - o Achieved a 40% reduction in objective value in contextual optimization and at least a 20% reduction in regret in control systems; selected as one of the 40 oral presentations at AISTATS 2024 from over 10,000 submissions.

Publications

- **Rayan, S.,** Patel, Y., Tewari, A. “A Greedy PDE Router for Blending Neural Operators and Classical Methods.” (In Submission) <https://arxiv.org/pdf/2509.24814>
- **Rayan, S.,** Tewari, A. “Learning to Partially Defer for Sequences.” (In Submission) <https://arxiv.org/pdf/2502.01459>
- Patel, Y., **Rayan, S.,** Tewari, A. “Conformal Robust Control of Linear Systems.” AISTATS 2026 <https://arxiv.org/pdf/2405.16250>
- Patel, Y., **Rayan, S.,** Tewari, A. “Conformal Contextual Robust Optimization.” AISTATS 2024 <https://proceedings.mlr.press/v238/patel24a.html>

Skills

Programming Languages & Tools: Python, C++, C, SQL, Java, R, Git, Docker, Linux, Vim, Gurobi
Machine Learning & AI: PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face transformers, LoRA, reinforcement learning, supervised fine-tuning.

Data Science & Analytics: NumPy, SciPy, Pandas, Jupyter, Tableau, Power BI

Other Experiences

University of Michigan, Ann Arbor, MI

Aug 2022 – Present

Graduate Student Instructor

- Instructed 40+ students per semester in R and Python for statistical methods, hypothesis testing, and deep learning, while grading assignments for 80 – 120 students in Intro to Statistics, Statistics and AI, and Bayesian Data Analysis.
- Created customized lab materials and practice problems to enhance students’ homework and exam readiness, earning an average teaching evaluation score of 4.5/5.

University of California, Berkeley, CA

May 2021 – May 2022

Undergraduate Research Fellow (Advised by Dr. Banghua Zhu and Dr. Jiantao Jiao)

- Designed a novel replay-based continual learning algorithm that effectively minimizes regret without catastrophic forgetting.
- Improved performance over state-of-the-art continual algorithms, reducing regret by 10%.